A Statistical Analysis of Randomness in Prime Gap Sequences

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Abstract

Prime numbers form a deterministic sequence, yet the gaps between consecutive primes fluctuate in a way that appears irregular. This study evaluates whether normalized prime gaps exhibit statistical signatures of randomness by analyzing primes up to 500,000 generated using the Sieve of Eratosthenes. Gaps were normalized by the logarithmic growth predicted by the Prime Number Theorem and compared against exponential random gaps and pseudorandom gaps from a linear congruential generator. Statistical tests revealed that while prime gaps show near-zero autocorrelation, they deviate significantly in distribution and entropy from both random and pseudorandom models. These results indicate that prime gaps exhibit partial randomness while retaining a detectable deterministic structure.

1 Introduction

Prime numbers are fundamental to number theory, yet their distribution contains irregularities that suggest a connection to randomness [4, 5]. The differences between consecutive primes—prime gaps—grow on average but fluctuate unpredictably, creating the appearance of random behavior within a deterministic system [3]. However, irregularity alone does not imply true randomness. Deterministic systems such as Rule 30 cellular automata, chaotic maps, and cryptographic hash functions demonstrate that simple rules can produce outputs that mimic randomness.

The Prime Number Theorem predicts that the average gap near a prime p is approximately $\log(p)$, but it does not explain individual fluctuations [3]. Cramér's probabilistic model proposes that primes behave like independent random events, implying that normalized prime gaps should follow an exponential distribution [1, 2]. This study evaluates the statistical behavior of prime gaps using distributional tests, entropy analysis, and autocorrelation measurements.

2 Background

While the Prime Number Theorem describes long-term trends, short-term fluctuations of prime gaps remain highly irregular [3]. One approach to understanding this behavior is Cramér's model, which treats primes as occurring independently with probability $1/\log(n)$ [1]. Under this assumption, normalized gaps are expected to follow an exponential distribution, similar to radioactive decay [2].

True randomness is defined by several distinct characteristics [8]:

- **Distribution:** Measures how values are spread. For example, rolling a fair die many times should

result in each value being equally likely to appear.

- **Independence:** Determines if values affect others. For example, a coin flip resulting in heads does not change the outcome of the next flip.

- **Unpredictability:** Measures the difficulty of guessing the next value. A sequence that repeats a pattern is highly predictable.

3 Method

3.1 Prime Generation and Normalization

Primes up to 500,000 were generated using the Sieve of Eratosthenes [3]. Prime gaps were computed as $g_n = p_{n+1} - p_n$. To account for growth trends, gaps were normalized:

$$g'_n = \frac{g_n}{\log(p_n)} \tag{1}$$

3.2 Comparative Models

To evaluate prime gaps, they were compared with two sequences:

- **Random Model:** Gaps generated using an exponential distribution.

- **Pseudorandom Model:** Generated using a linear congruential generator (LCG): $X_{n+1} \equiv (aX_n + c) \pmod{m}$.

3.3 Statistical Tests

Evaluating randomness requires checking several independent properties [8]:

- **Chi-Square (χ^2):** Measures fit to exponential distribution:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \tag{2}$$

- **Kolmogorov–Smirnov (KS):** Compares the empirical distribution $F_n(x)$ to the theoretical model $F(x)$:

$$D = \sup_x |F_n(x) - F(x)| \tag{3}$$

- **Shannon Entropy (H):** Quantifies unpredictability and information density [9]:

$$H = -\sum p_i \log(p_i) \tag{4}$$

- **Autocorrelation (ρ):** Measures local independence between consecutive gaps.

4 Results and Discussion

Table 1: Full Statistical Test Results for Primes up to 500k

Metric	Seq.	100k	200k	300k	400k	500k
χ^2	P vs R	5530	6080	8200	9020	6730
	P vs Ps	3440	3600	11500	3280	1380
KS Stat	P vs R	0.185	0.184	0.180	0.175	0.174
	P vs Ps	0.419	0.397	0.412	0.425	0.430
Entropy	Prime	3.59	3.69	3.75	3.79	3.82
	Random	13.2	14.1	14.7	15.0	15.3
	Pseudo	13.2	14.1	14.7	15.0	15.3
Auto-ρ	Prime	-0.070	-0.064	-0.050	-0.044	-0.042
	Random	-0.007	0.009	0.007	-0.004	-0.0003
	Pseudo	-0.001	0.006	0.005	0.005	0.004

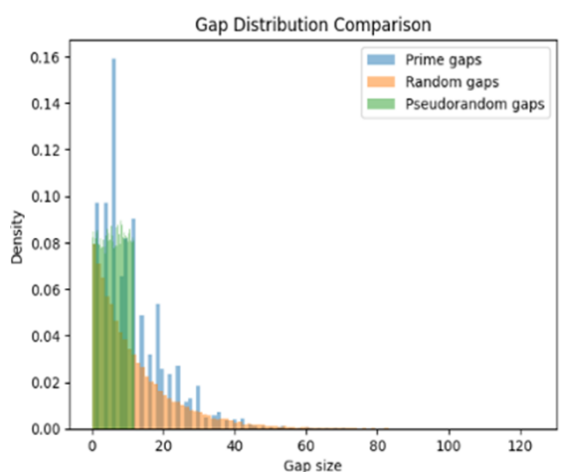

Figure 1: Gap Distribution Comparison (Chi-Square).

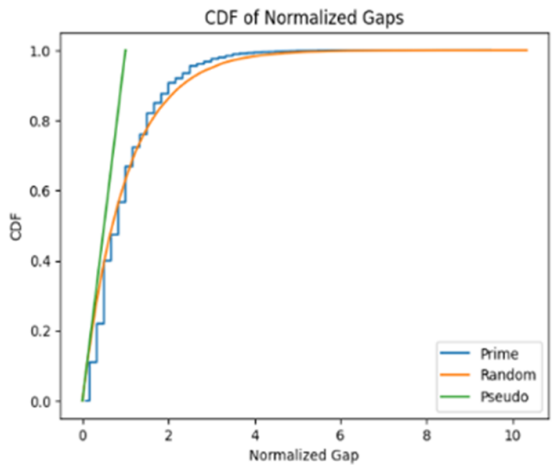

Figure 2: Kolmogorov–Smirnov Cumulative Distribution Plot.

The chi-square test showed significant deviation from the exponential distribution. The KS test supported this, showing larger differences for prime gaps than in comparison models [7]. Entropy analysis revealed prime gaps have noticeably lower entropy, indicating greater internal structure [9]. Autocorrelation values were close to zero, matching the independence of random models.

5 Conclusion

This study demonstrates that prime gaps exhibit a blend of random-like and deterministic behavior. Their near-zero autocorrelation suggests independence, but significant deviations in chi-square and KS test indicates that the overall shape of the distribution differs from the exponential model. Lower entropy further distinguishes prime gaps from truly random data.

These findings together agree with the hypothesis that prime gaps deviate from true randomness, especially in unpredictability and distribution. This distinction between prime gaps and randomness is particularly important in the aspect of modern cryptography. Many modern cryptographic systems rely on the unpredictability of prime gaps to maximize the strength of the algorithm. These finding clarify the limits of treating prime derived sequences as random and helps clarify assumptions.

Future work may involve applying the full NIST randomness test suite, extending the analysis to larger primes, or comparing results to more advanced probabilistic models such as Cramér–Granville refinements or predictions from random matrix theory.

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